4th International Conference on Smart Sustainable Development

SDG Projects — Application Form

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1. Title of Project

Road Accident Risk Predictor: Explainable Risk Scoring for Infrastructure Safety Prioritization

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2. Team members

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5. Summary (400–500 words)

Road safety agencies operate with finite resources for inspection, signage, lighting, and speed-limit review, but road trauma is not distributed evenly across a network. It concentrates in particular combinations of road and environmental conditions. The Road Accident Risk Predictor addresses a specific, practical question: given limited resources, which road conditions should be prioritized first?

The project is a machine learning system built on a Random Forest Regressor trained on 517,754 road-condition scenarios, using 15 features including three engineered interaction terms (speed times curvature, accidents per lane, and a night-plus-bad-weather flag). The model predicts a continuous risk score for a given combination of road type, curvature, speed limit, lighting, weather, signage, and traffic context, and explains every prediction using SHAP (SHapley Additive exPlanations) feature attribution, so the output is not a

black-box number but an interpretable breakdown of which factors are driving the score up or down for that specific scenario.

The system is live and publicly deployed as an interactive web application, where a user can adjust road and environmental conditions and see the predicted risk update in real time, alongside the SHAP explanation and context-aware safety recommendations. Validation performance is R-squared 0.88, independently reproduced rather than only self-reported, with 5-fold cross-validation stability of plus-or-minus 0.0003. The model was checked for data leakage (no duplicate identifiers, and the dataset has no geographic or temporal fields that a random train/test split could leak across) and for subgroup performance stability across road type, weather, and lighting conditions, with results and limitations documented transparently rather than only headline metrics reported.

The core evidence behind the project's value is a risk-concentration result computed directly from the model's own output: ranking all 517,754 training scenarios by predicted risk, the highest-risk 10 percent of conditions carry 1.93 times their proportional share of total predicted risk. This demonstrates that the model finds genuine, usable concentration rather than noise, which is the precondition for any targeting strategy to be worth pursuing. Building on this, the project sets out a scenario-based estimate of potential benefit under stated intervention-effectiveness assumptions drawn from published road-engineering literature, explicit about what it does and does not claim: the model has not been shown to prevent any specific crash, and the estimate is a targeting-value illustration, not a causal claim.

The project also documents its own boundaries directly: it uses road and environmental data only, with no driver identity or demographic information, and is intended for infrastructure planning and safety prioritization, not individual driver liability or profiling. It is described honestly as a live decision-support prototype rather than a production system with an operator, and known limitations, including reduced precision in low-light conditions and the synthetic nature of the training data, are disclosed rather than omitted. The result is a system whose primary contribution is not a single accuracy number but a demonstrated, explainable, and honestly bounded way of turning road condition data into prioritization decisions.

6. Impact (100–150 words)

The project's primary impact evidence is measured directly from the model, not assumed: the highest-risk 10 percent of road conditions in the training data carry 1.93 times their proportional share of total predicted risk, meaning limited safety-intervention resources can be concentrated on a small share of conditions for disproportionate expected benefit, rather than spread evenly across a network. Under scenario-based intervention-effectiveness assumptions (15 to 30 percent, sourced from the FHWA Crash Modification Factors Clearinghouse), this targeting approach represents a measurable potential benefit for road safety agencies, insurers, and the public who share the road. The project is explicit that this is a scenario-based potential impact and not a causal claim of prevented harm, since demonstrating that would require a deployed intervention study this project has not conducted.

7. Alignment with the UN's Sustainable Development Goals — SDGs (200 words)

The project targets SDG 3.6 as its primary goal: halve global deaths and injuries from road traffic accidents. The mechanism is direct: the model's risk-concentration result shows that predicted risk is not evenly distributed across road conditions, which means targeted infrastructure spending, rather than evenly spread investment, can plausibly achieve a larger reduction in serious injuries for the same resource outlay. The secondary goal is SDG 11.2, safe, affordable, and sustainable transport systems, since the model's explicit purpose is supporting infrastructure planning decisions such as signage, lighting, and speed-limit review, which are the concrete interventions that make transport systems safer for all road users.

Two supporting targets were deliberately not claimed. An earlier draft of this project included SDG 9.c, but this was removed after review because 9.c concerns ICT and internet access in least-developed countries specifically, which does not fit this project's mechanism. The project favors precision over a broad SDG list, on the view that a small number of directly justified goals is stronger evidence than a longer, weaker list.

8. Other Comments

The live demo (https://roadaccident-roshanar-aryal.streamlit.app/) is hosted on Streamlit Community Cloud's free tier, which sleeps after a period of inactivity. Cold start was measured directly during preparation of this submission at approximately 95 seconds from click to fully rendered app. If the link shows "this app has gone to sleep," clicking "Yes, get this app back up!" and waiting roughly a minute will resolve it. Screenshots of the live application and a walkthrough recording are included with this submission as a fallback in case the reviewer encounters the app mid-wake.

Full supporting technical detail beyond what fits in the word limits above — the complete model validation and leakage audit, the full risk-concentration table, and the complete responsible-AI and limitations documentation — is included in the supporting materials.

9. A URL / web link for a shared folder of supporting documents

GitHub repository: https://github.com/roshanaryal1/SafeRoute

(Full evidence packet — screenshots, walkthrough video, model validation, impact analysis, responsible AI documentation — submitted as a zip file alongside this form via both the SSD2026 Google Form portal and as an email attachment to info@smartsust.org, per the conference's stated submission instructions.)